Intelligent Systems – Knowledge Representation

SCJ3553
Artificial Intelligence

Faculty of Computer Science and Information Systems
Universiti Teknologi Malaysia

Outline

- Introduction
- Semantic Network
- Frame
- Conceptual Graph
- Agent-based

Introduction

- Implementation of intelligent systems
  - Representation scheme (like data structures, explicit structure for knowledge representation)
  - Representation medium (like programming languages, i.e. PROLOG, LISP,C++,Java, JSP, ASP.NET,PHP….)

Semantic Network

- Graph- explicitly representing relations using arcs and nodes; formalizing knowledge
- Semantic network
  - represents knowledge as a graph
  - with nodes corresponding to facts or concepts
  - and arcs correspond to relations between concepts
  - Eg. Figure 1(canary-bird) and Figure 2(snow-ice)

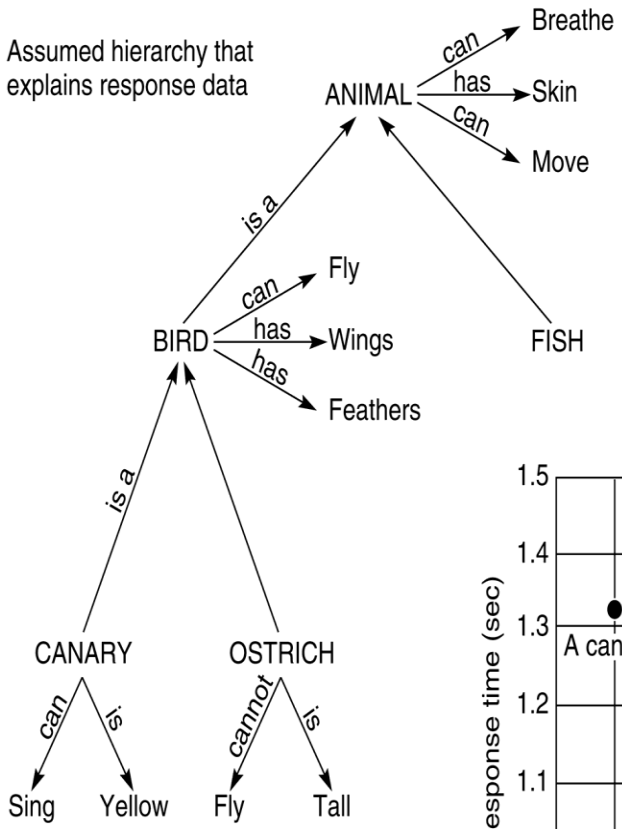
Semantic Network

- Knowledge organization
- Inheritance systems allow:-

-storing knowledge at highest level of abstraction

-reduce size of knowledge base

-help prevents update inconsistencies



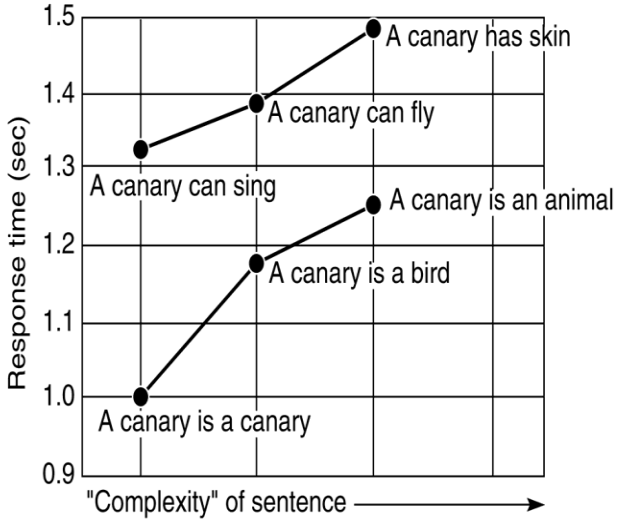


Figure 1: Semantic network developed by Collins and Quillian in their research on human information storage and response times (Harmon and King 1985).

Semantic Network

- Semantic network can be used to:-

-answer questions about snow,ice and snowmen(with appropriate inference rules)

-References are made by following links to concept

-implement inheritance i.e. frosty inherits all properties of snowman

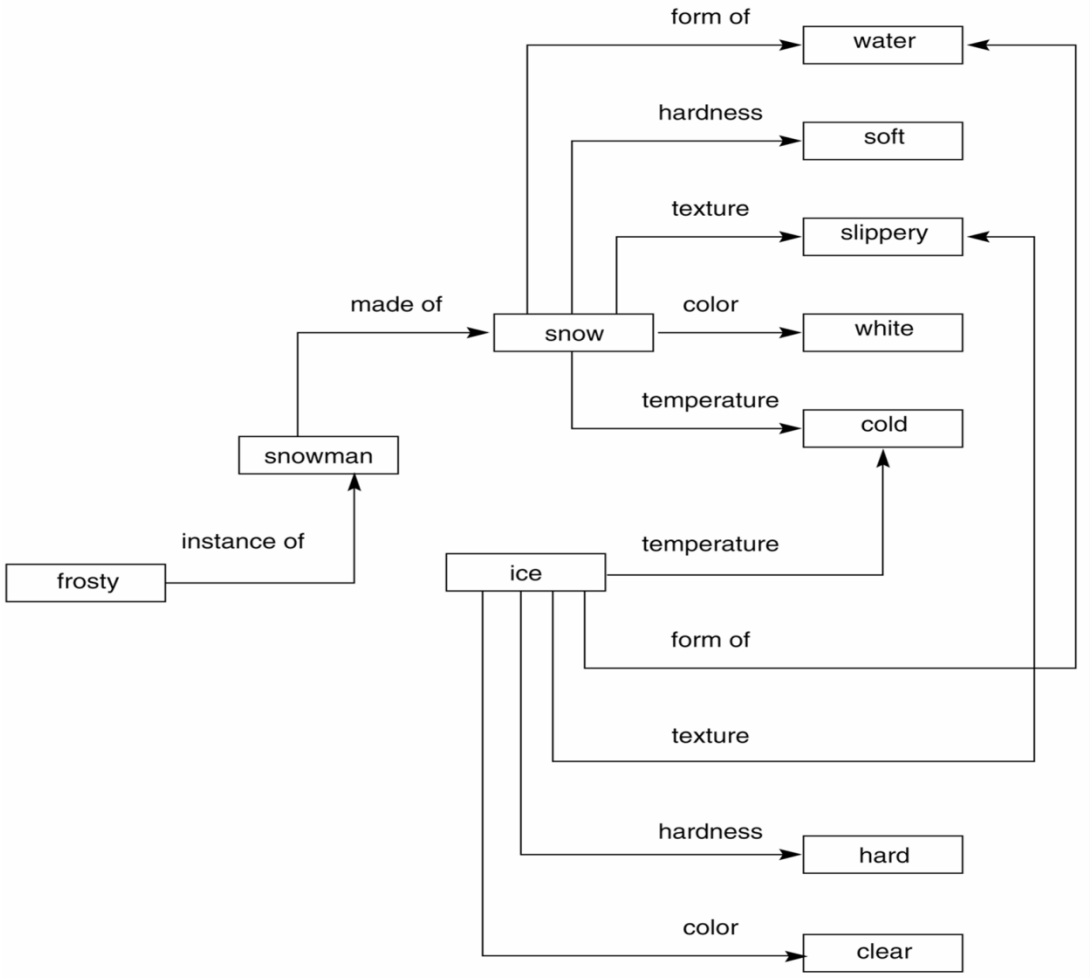


Figure 2: Network representation of properties of snow and ice

Semantic Network

- A program defined English word
- Each definition leads to other definition in unstructured or circular fashion
- Looking up a word, the network is traversed until the word is understood
- Knowledge-based is organized into planes, each explains single word
- i.e. 3 planes capture definitions of ‘plant’-living organism,work place or putting seed in ground

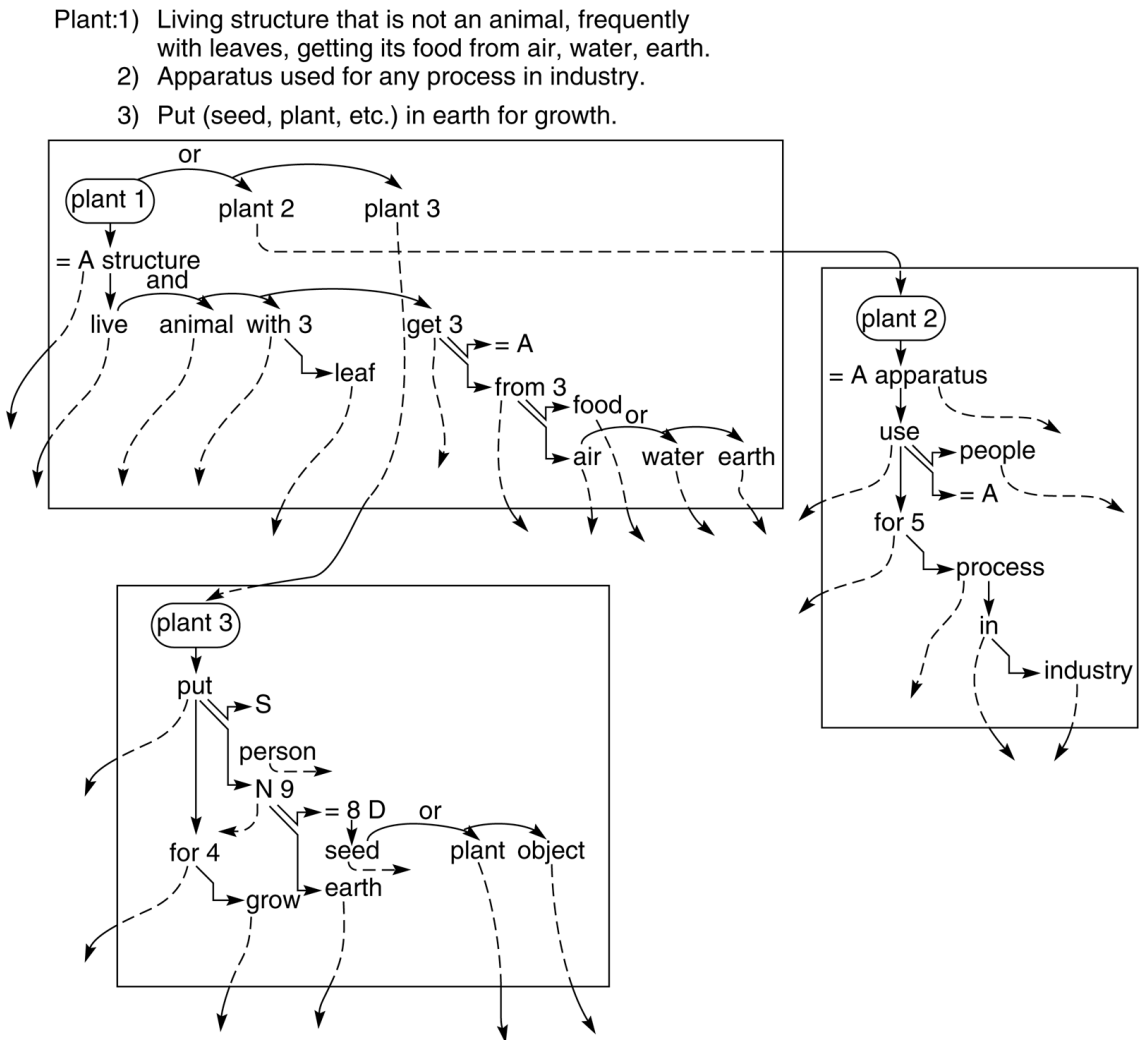


Figure 3: Three planes representing three definitions of the word “plant” (Quillian 1967)

Semantic Network

- Knowledge-based is used to find relationships between pairs of english word

- Given two words, it would search graphs outward from each word in breadth-first fashion, searching for **intersection-node**

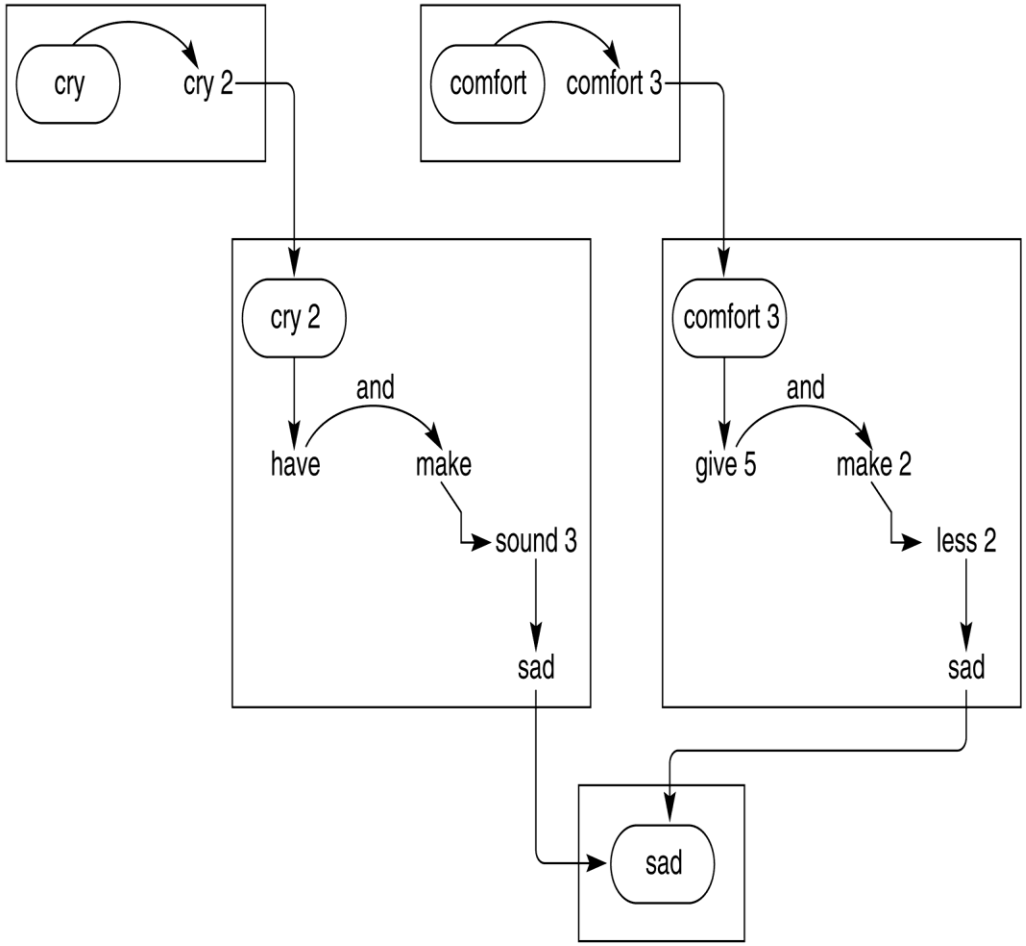
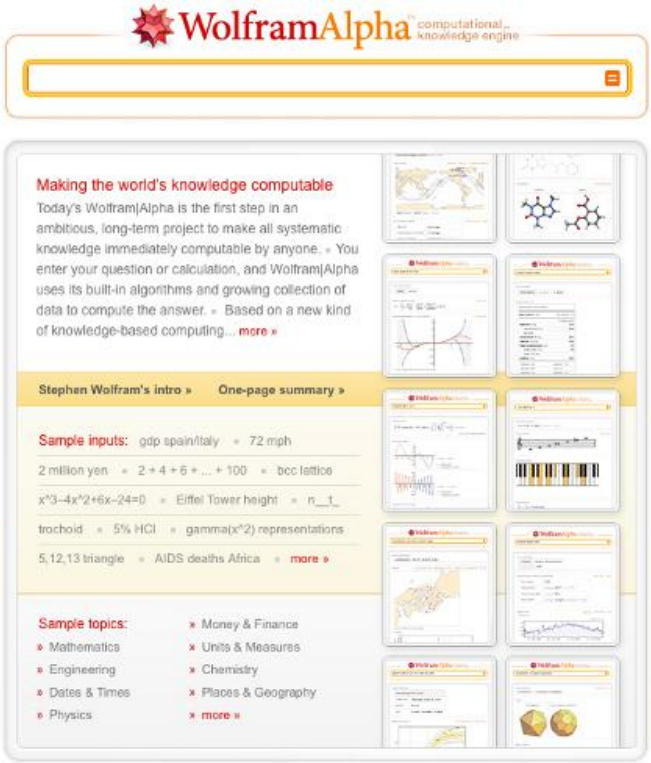


Figure 4: Intersection path between “cry” and “comfort” (Quillian 1967).

Semantic Network







Frame

- Another representational scheme
- Implicit connection of information
- Static data structure to represent well-understood stereotyped situations
- Organize our knowledge of the world
- We adjust to new situation by calling up information structure by past experience
- We then revise details of past experiences to represent differences in new situation

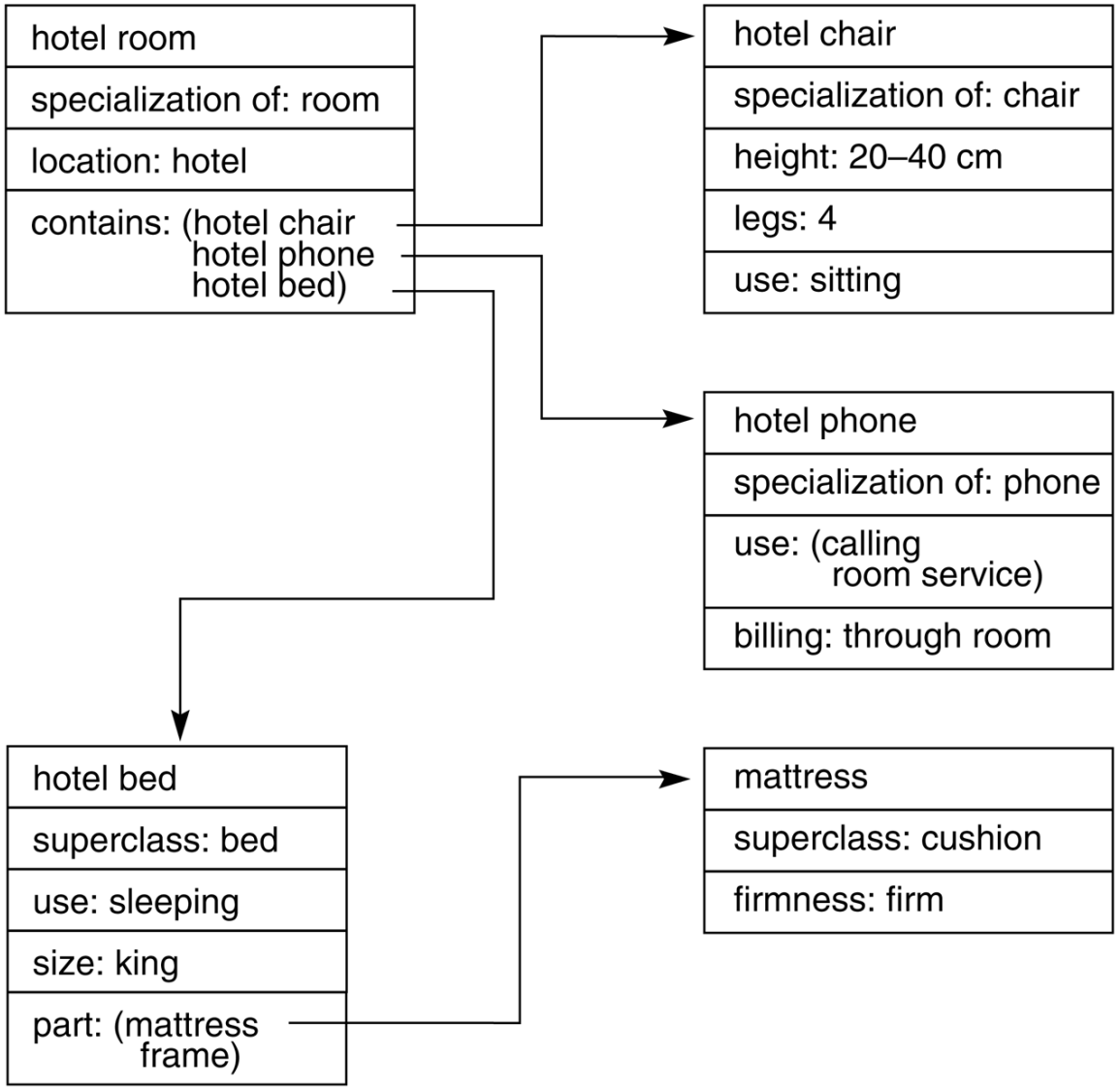
Frame

A frame includes:

1. *Frame identification information*.
2. *Relationship of this frame to other frames*. The "hotel phone" might be a special instance of "phone," which might be an instance of a "communication device."
3. *Descriptors of requirements for a frame.* A chair, for instance, has its seat between 20 and 40 cm from the floor, its back higher than 60 cm, etc. These requirements may be used to determine when new objects fit the stereotype defined by the frame.
4. *Procedural information on use of the structure described*. An important feature of frames is the ability to attach procedural code to a slot.
5. *Frame default information*. These are slot values that are taken to be true when no evidence to the contrary has been found. For instance, chairs have four legs, telephones are pushbutton, or hotel beds are made by the staff.
6. *New instance information*. Many frame slots may be left unspecified until given a value for a particular instance or when they are needed for some aspect of problem solving. For example, the color of the bedspread may be left unspecified.

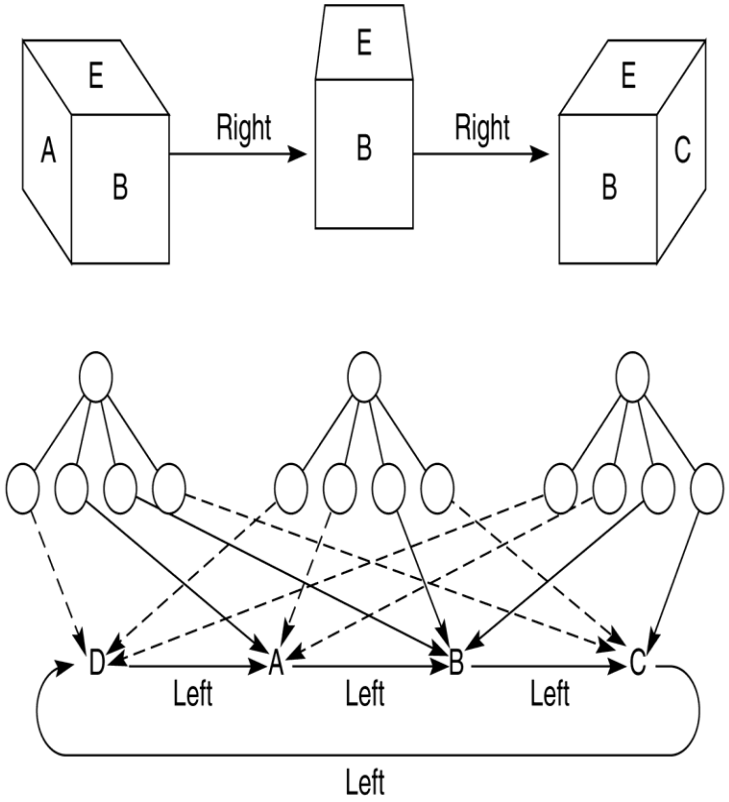
Frame

- Hotel room and its components are described by number of individual frames
- Each frame may be seen as data structure
- Contains info relevant to stereotyped entities
- Frame systems support class inheritance



Frame

- This frame system represents four of faces of cube
- Broken line indicates face out of view from that perspective
- Links between frames indicate relations between views represented by frames
- Each slot in one frame could be a pointer to another entire frame
- Since given information can fill many different slot (face E), No redundancy in information stored
- Frames allow complex objects to be represented as a single frame rather than large network structure

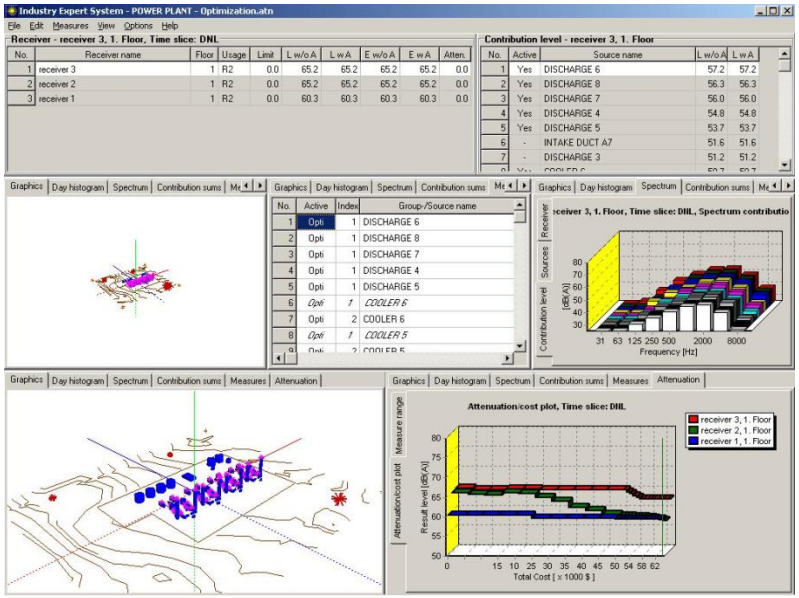


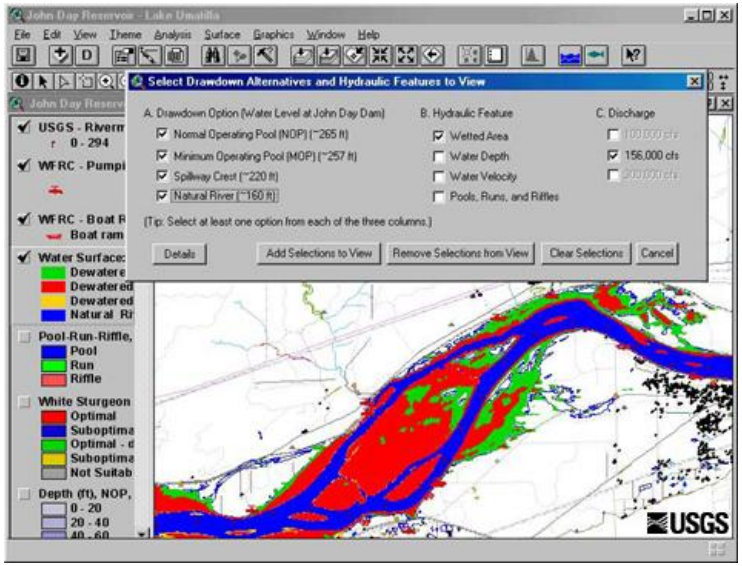
Spatial frame

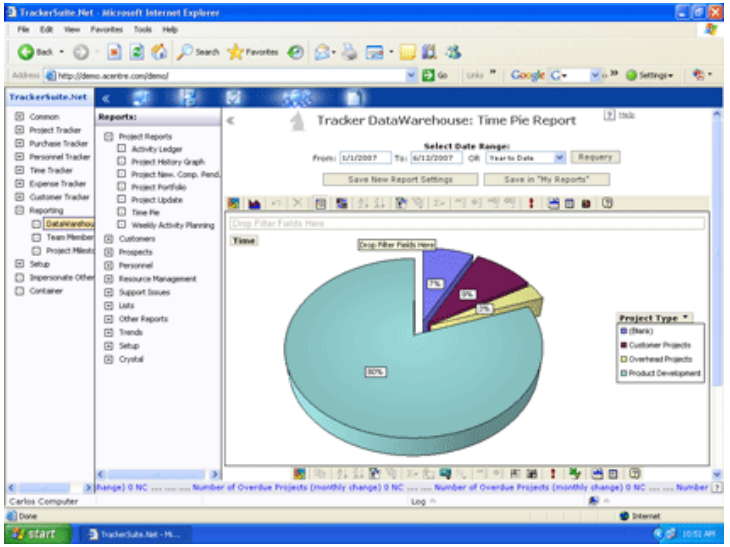
Pictorial frames

Relation markers in common-terminal structure can represent more invariant (e.g., three-dimensional) properties

Frame


Industry Expert System - POWER PLANT - Optimization.atn
Receiver - receiver 3, 1. Floor, Time slice: DNL
Contribution level - receiver 3, 1. Floor
Attenuation/cost plot, Time slice: DNL
Total Cost [x 1000 $]


John Day Reservoir - Lake Umatilla
Select Drawdown Alternatives and Hydraulic Features to View
A. Drawdown Option (Water Level at John Day Dam)
B. Hydraulic Feature
C. Discharge
Details
Add Selections to View
Remove Selections from View
Clear Selections
Cancel
USGS

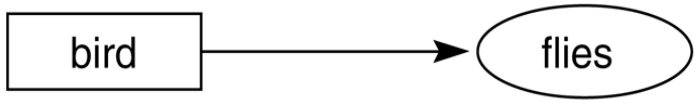

TrackerSuite.Net - Microsoft Internet Explorer
Tracker DataWarehouse: Time Pie Report

Conceptual Graph

- Example of a network representation language
- A finite, connected graph
- Nodes are either:
  - concepts or
  - conceptual relations
  - No labeled arcs
  - Conceptual relation nodes=relations between concepts
  - i.e. (dog and brown ~ concept nodes)
  - i.e. (color ~ conceptual relation)

Conceptual Graph

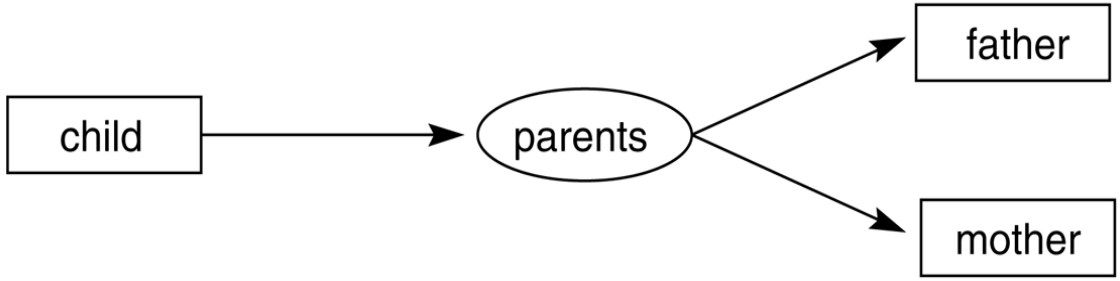
- Each conceptual graph represents a single proposition
- A typical knowledge-base contains several graphs
- Graph must be finite
- i.e. a dog has a color of brown
- Conceptual graphs are used to model semantics of natural language



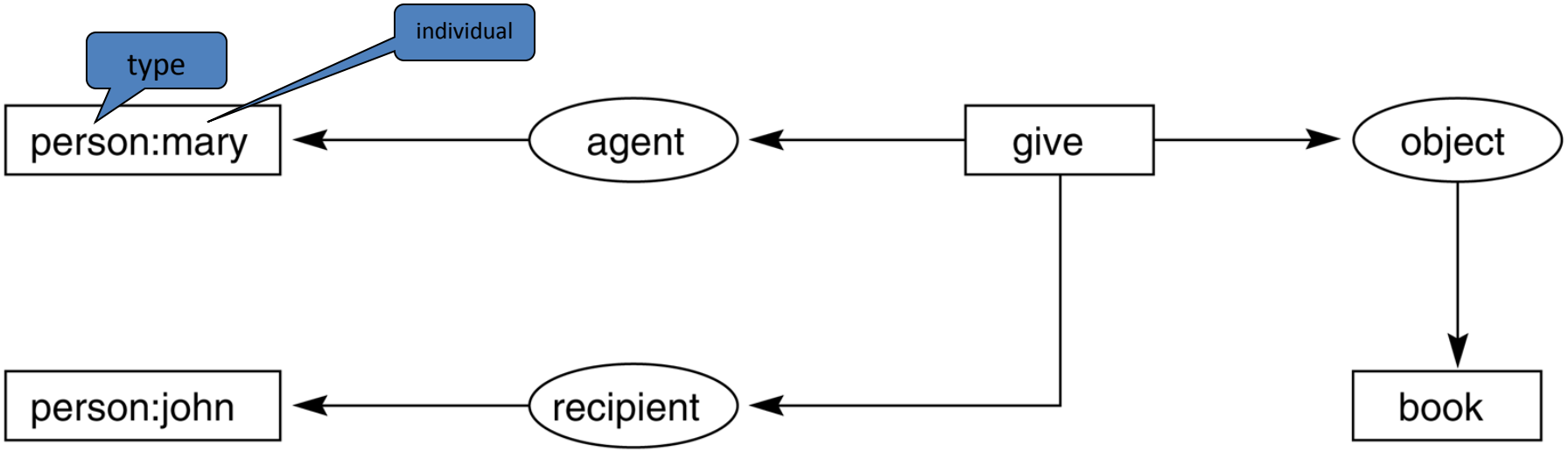
Flies is a 1-ary relation.



Color is a 2-ary relation.



Conceptual Graph



- The graph uses conceptual relations to represent cases of the verb ‘to give’
- Conceptual graphs used to model semantics of natural language
- Every concept is a unique individual of a particular type, separated by :

Conceptual Graph

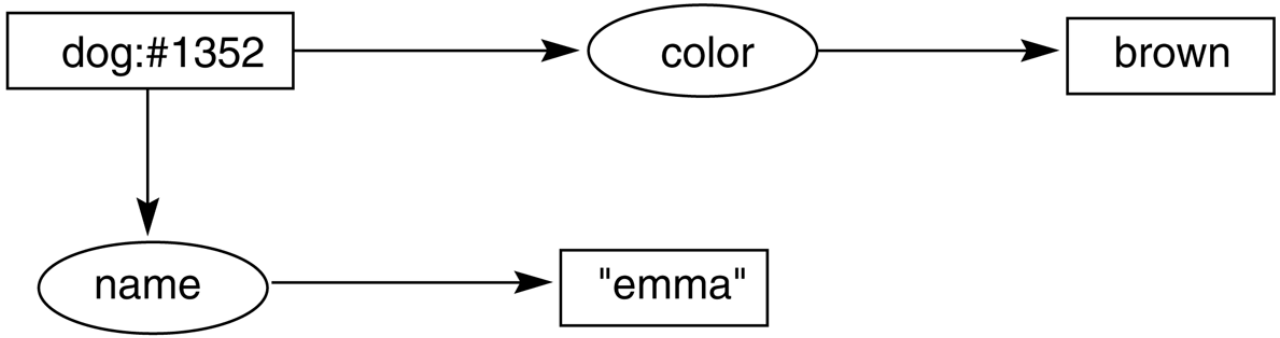
Conceptual graph indicating that the dog named emma is brown.



Conceptual graph indicating that a particular (but unnamed) dog is brown.



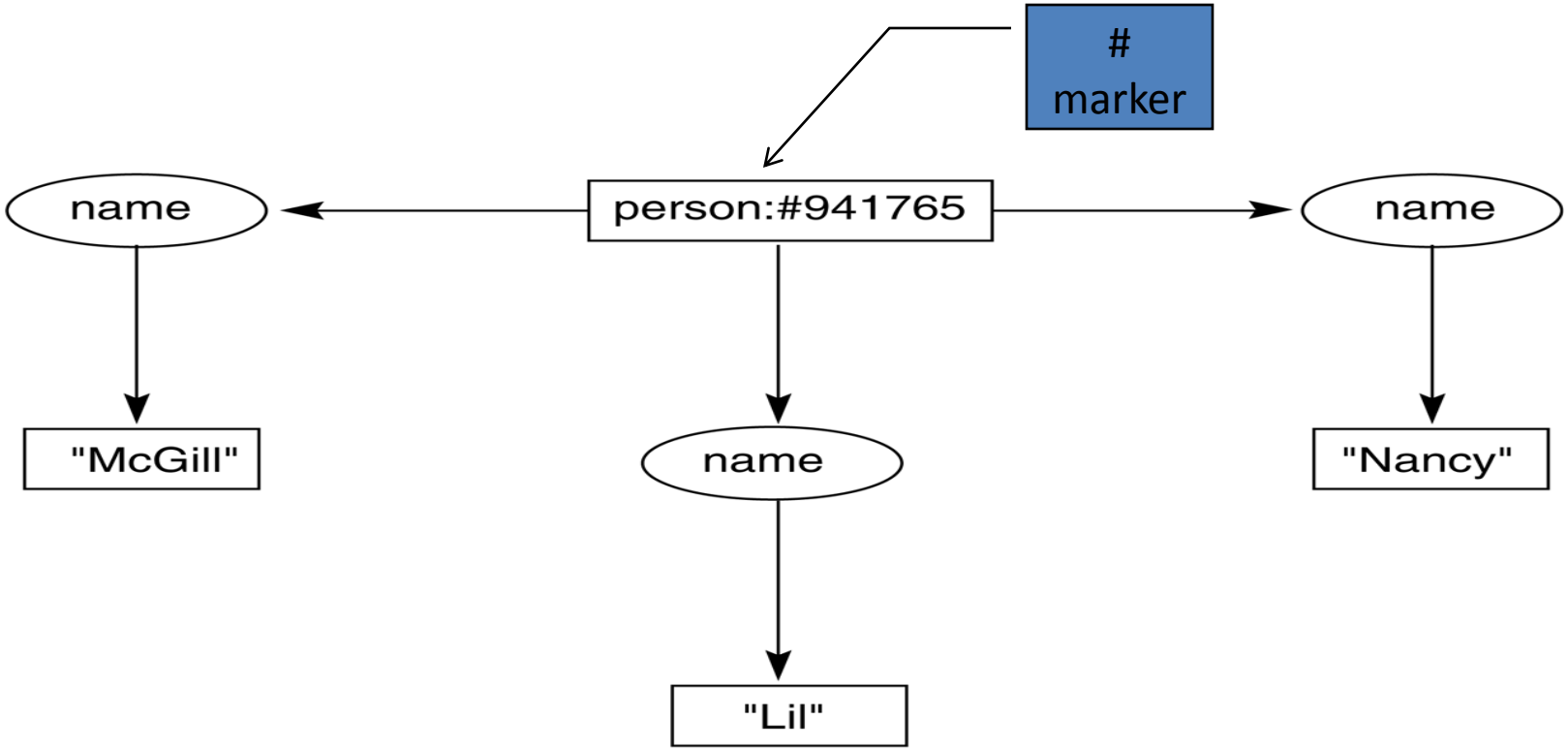
Conceptual graph indicating that a dog named emma is brown.



Graph
1= 3

Conceptual Graph

Conceptual graph of a person with three names.

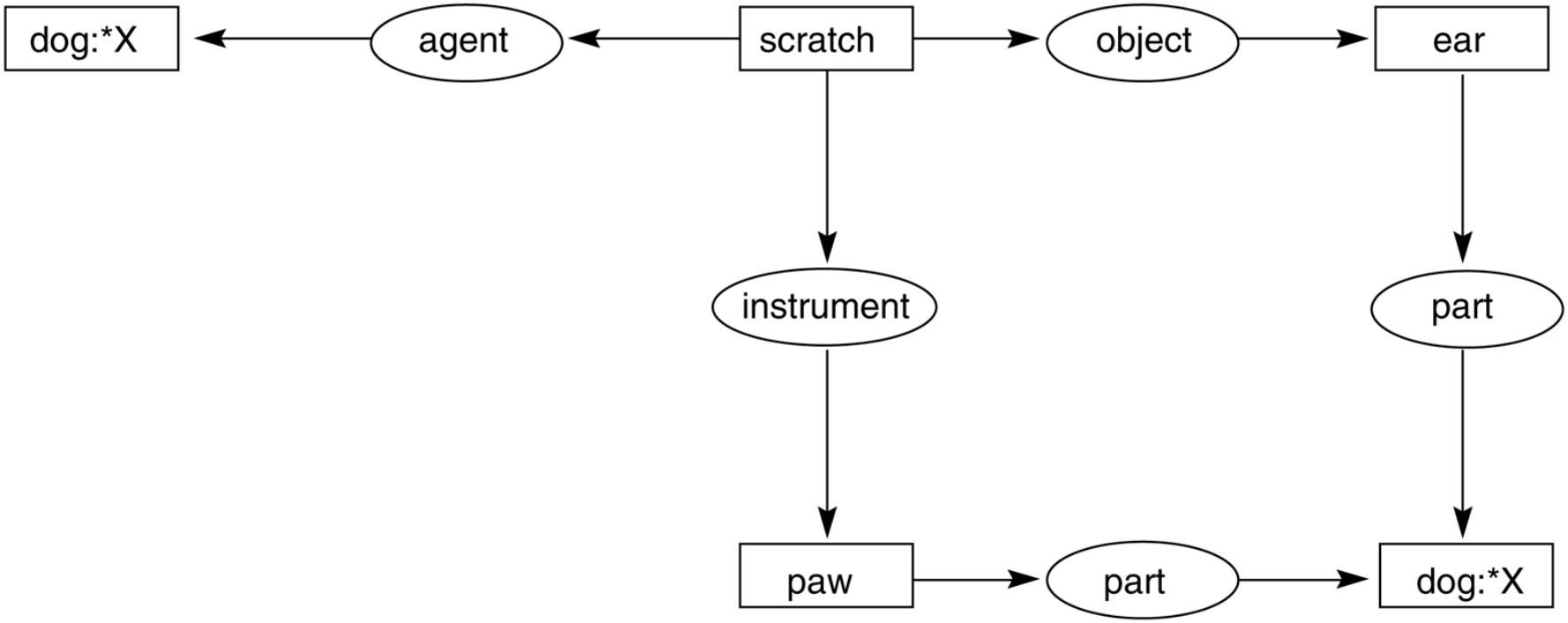


Marker

- is unique and different from names
- Individual has many names but one marker
- different individuals may have same name, but not same marker

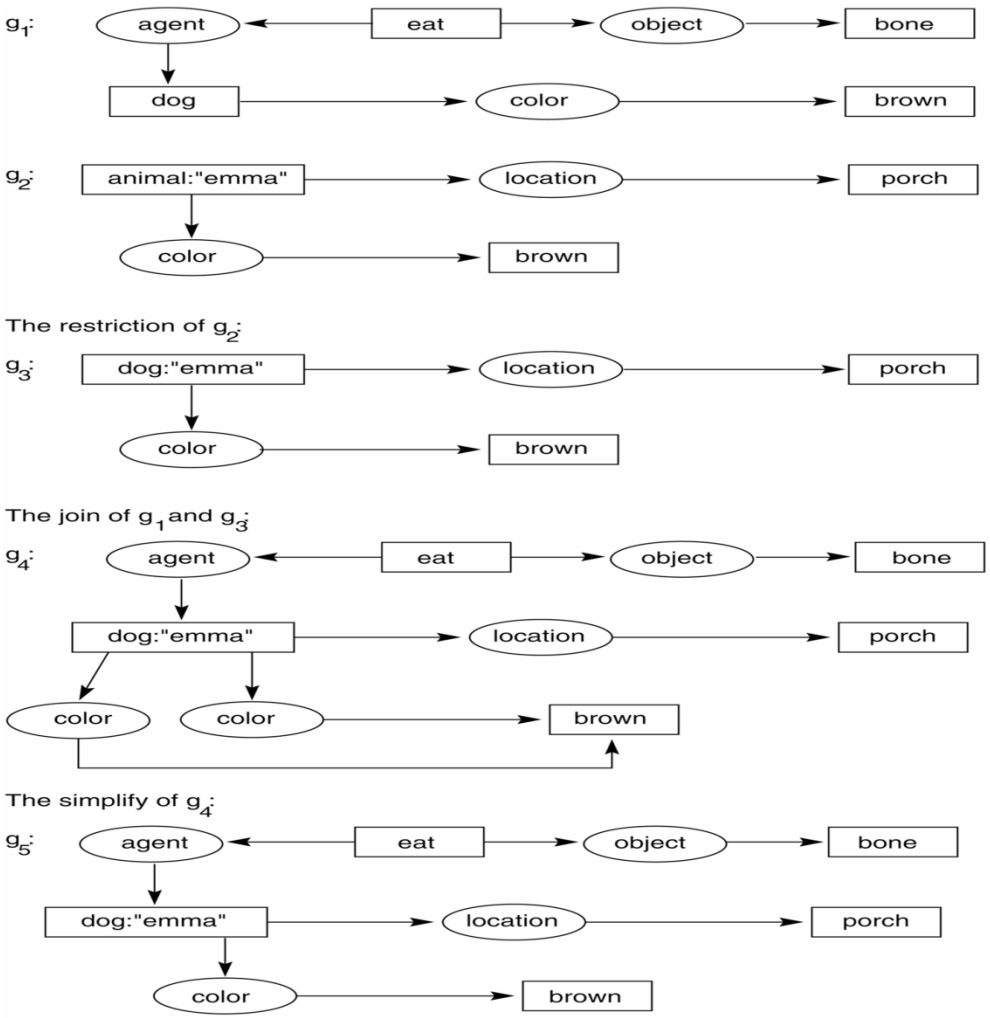
Conceptual Graph

Conceptual graph of the sentence “The dog scratches its ear with its paw.”



- To summarize: each concept node indicates individual of specified type
- This individual is the referent of the concept
- Individual concept ~referent uses individual marker
- Generic concept ~ referent uses generic marker

Conceptual Graph



Conceptual graph includes operations creating new graphs from existing ones:

-specializing or generalizing existing graph to represent semantics of natural language:-

1)Copy-exact copy of graph

2)Restrict-replace concept nodes with specialized note:

- generic marker replace individual marker
- Replace type with subtypes; animal -> dog

Generalization and Specialization: Examples of restrict, join, and simplify operations

Conceptual Graph

3)Join- combine 2 graphs into single one

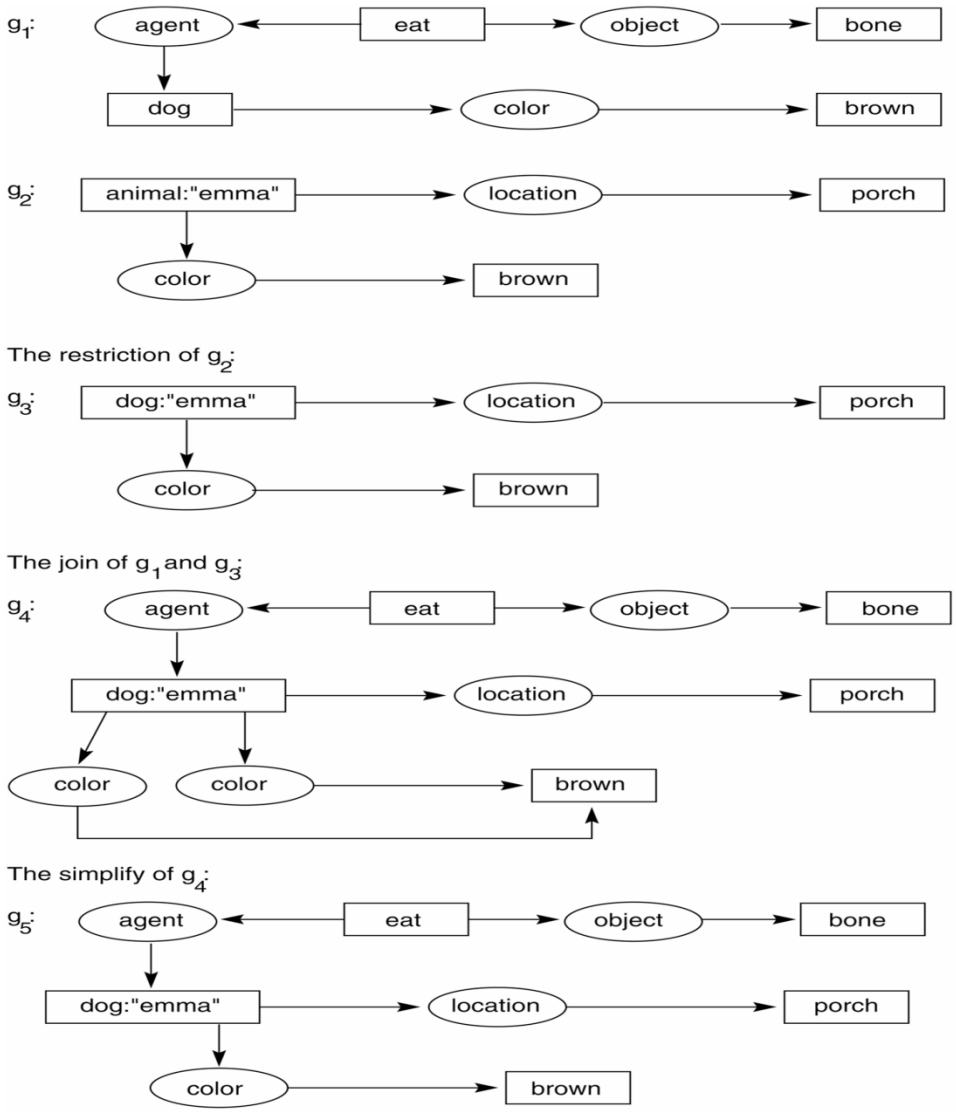
- If concept node c_1 and c_2 identical, delete c_2; c_1 replace c_2
- Specialization- produce less general graph

4)Simplify-if graph has duplicate relations, delete one together with its arcs

- Occur after join operation

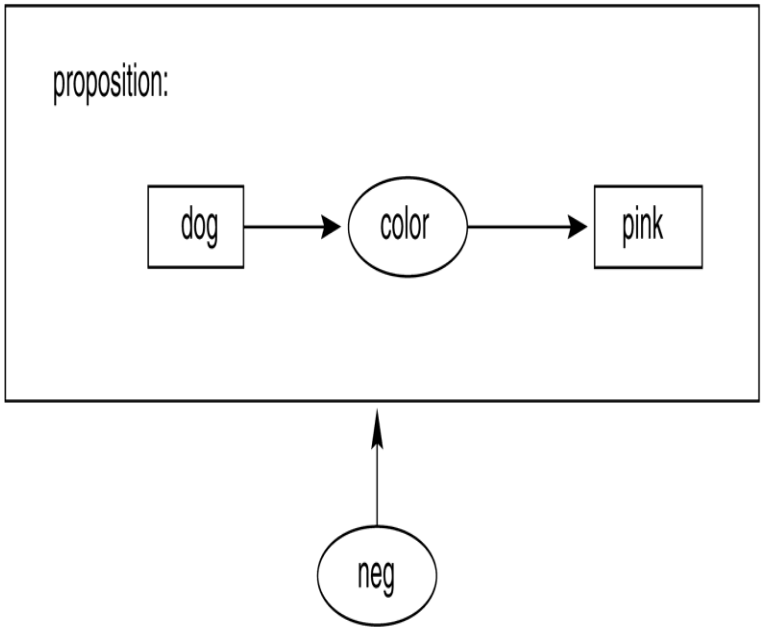
Ж restrict ~ match two concepts

Ж join and restrict ~ allow implementation of inheritance



Conceptual Graph

Conceptual graph of the proposition "There are no pink dogs."



- To represent negation or disjunction- variable quantification

  (universal quantifier $\forall$ *(for all) and* existential quantifier $\exists$ *(there exist)*

- neg – takes argument as proposition concept and assert that concept as false
- In conceptual graph, generic concepts are assumed to be existentially quantified
- Eg. Translations

$\exists$ X $\exists$ Y (dog(X) $\wedge$ color(X,Y) $\wedge$ brown(Y) ==> existential quantifier

$\forall$X $\forall$Y ($\neg$ (dog(X) $\wedge$ color(X,Y) $\wedge$ pink(Y)))

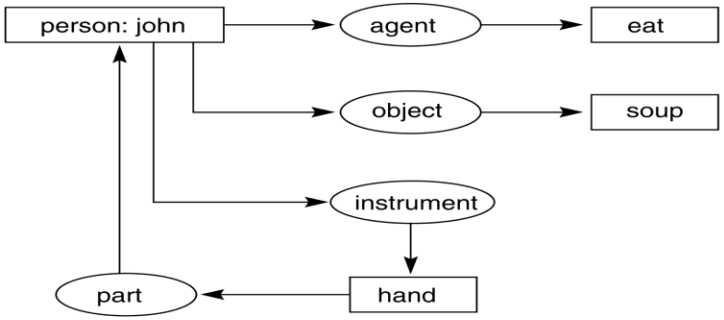
==> universal quantifier

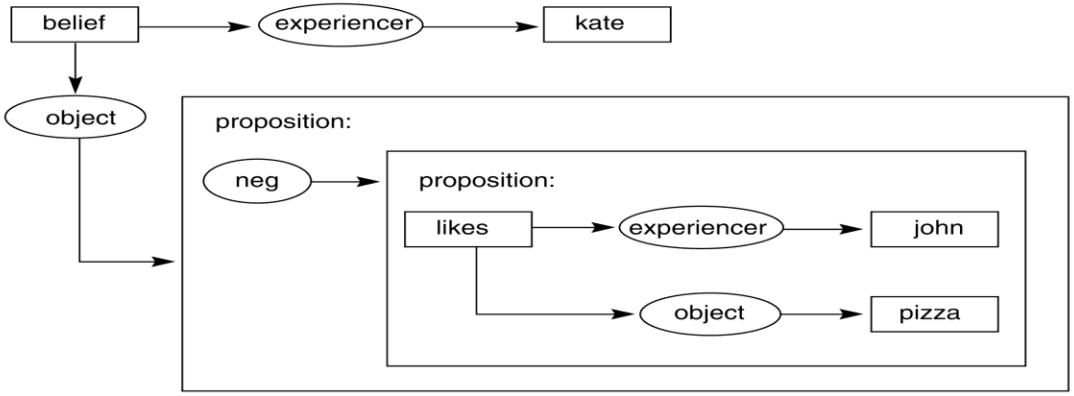
Eg. Translation

$\exists X_1$ (dog(emma) $\wedge$ color(emma, X_1) $\wedge$ *brown*(X_1))

- There is straight mapping from conceptual graph into predicate calculus notation (Sowa,1984)
- Advantage of conceptual graph – support some special-purpose inferencing mechanisms such as join and restrict, not normally part of predicate calculus

Conceptual Graph

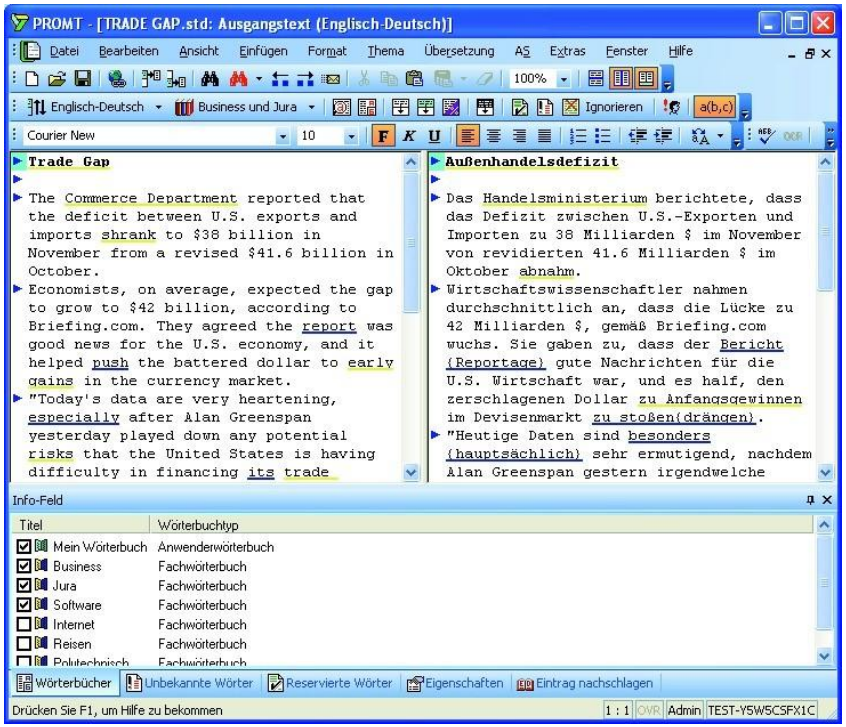


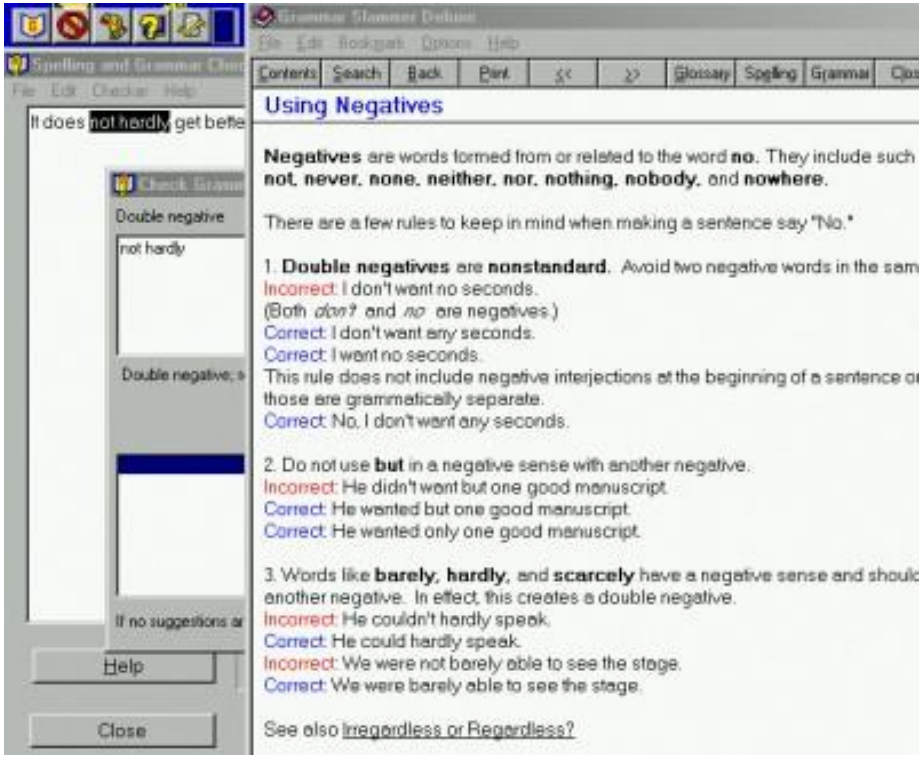


On a piece of paper , answer the questions below and submit

- Translate the two conceptual graphs into English
- Translate the two conceptual graphs into predicate calculus

Conceptual Graph

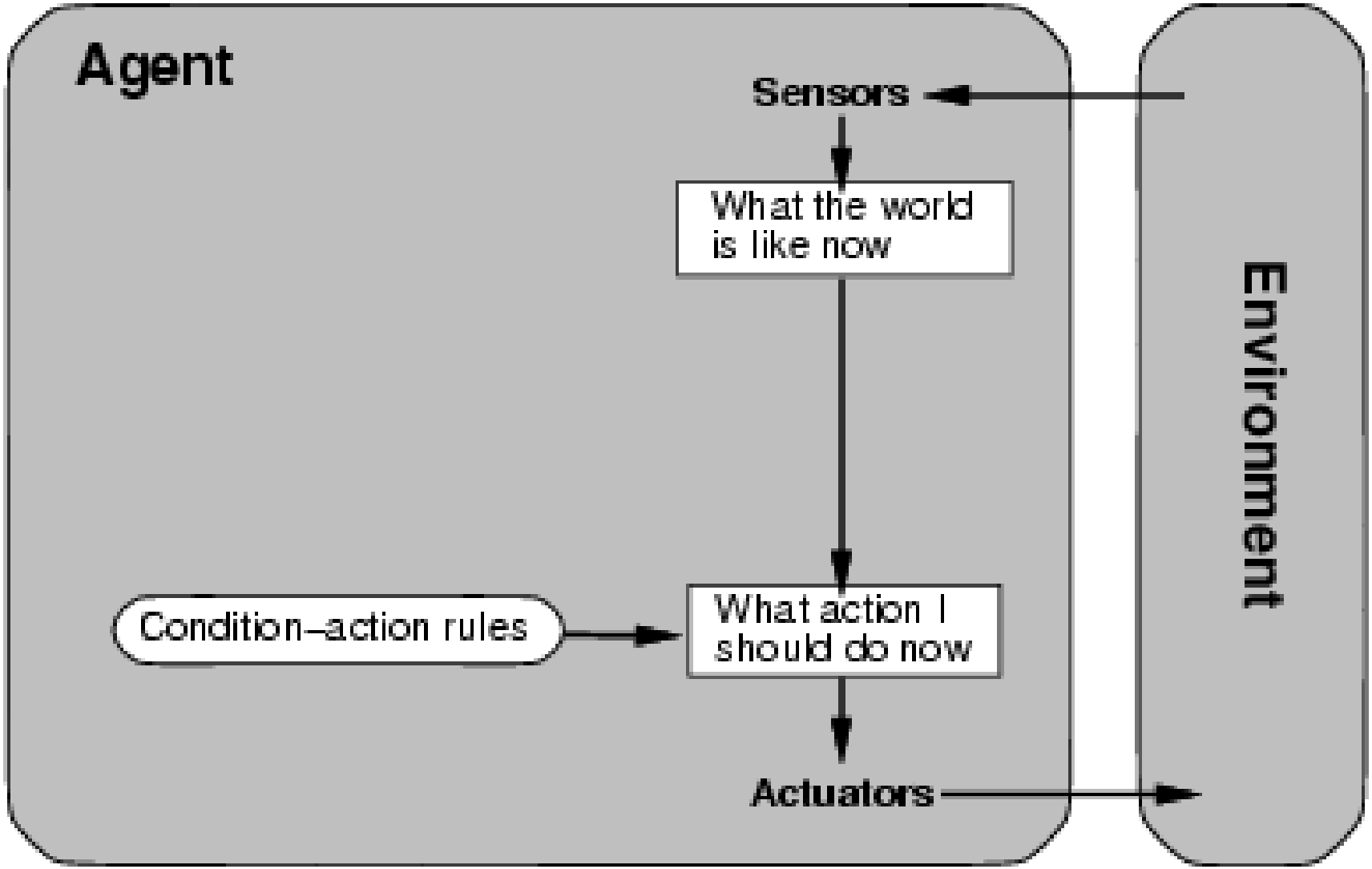




Agent

- Definition:
  - multi-agent :- a comp program with problem solvers situated in interactive environments, capable of flexible, autonomous and socially organized actions.
  - 4 criteria of intelligent agent:
    - Situated, autonomous, flexible and social

Agent


Agent
Sensors
What the world is like now
Condition–action rules
What action I should do now
Actuators
Environment

Agent

- Situatedness
  - Means agent receives input from environment in which it is active and can also effect changes within that environment
  - i.e. situations like internet, game playing, robotic situation
  - i.e. ROBOCUP competition – agent interact with ball and opponent

Agent

- Autonomous
  - Can interact with its environment without direct intervention of other agents
  - Control over its own action
  - Can also learn from experience to improve performance
  - i.e. ROBOCUP – agent pass the ball to a teammate or kick on goal depending on its situation

Agent

- Flexible
  - Intelligently responsive – receive stimuli from its environment and responds to them in an appropriate and timely fashion
  - proactive – not simply responsive but able to be opportunistic, goal directed and have appropriate alternatives for various situations
  - i.e. soccer agent –change its dribble depending on the challenge pattern of opponent

Agent

- Social
  - Interact with other software or human-agent towards the goal
  - social dimension address difficult situation
  - i.e. ROBOCUP – to score a goal, how one agent support another agent's goal?

Agent

- Multi-agent problem characteristics:
  1. Each agent has incomplete information and capabilities to solve entire problem
  2. No global system controller for entire problem solving
  3. Knowledge and input data for the problem is decentralized
  4. The reasoning process are asynchronous

Agent

- Agent VS Object
  - AGENT
    - request action to be performed
    - designed to have flexible
    - have own thread of controls
  - Object
    - invoke methods on one another
    - defined as computational systems with encapsulated state
    - have methods associated with state
    - communicate by message passing
    - rarely exhibit control over own behavior

Agent

